

Arnav Pundir

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer proficient in building high-performance, real-time, and offline-first web applications. Experienced in React 19, Next.js 15, Node.js, Python (Flask), TypeScript, RxDB, and WebRTC. Strong expertise in engineering low-latency systems, event-driven architectures, state synchronization, and delivering production-grade freelance client projects.

SKILLS

Languages : TypeScript, JavaScript, Python, C++, SQL, HTML/CSS
Frontend : React 19, Next.js 15, Tailwind CSS, Three.js, GSAP, Framer Motion, RxDB (PWA)
Backend & Systems : Node.js, Express.js, Flask, FastAPI, WebRTC, Redis, REST APIs, WebSockets
Databases & Cloud : Supabase (PostgreSQL / pgvector), MongoDB, Clerk Edge Auth, IndexedDB, Docker, Git

EDUCATION

COER University Roorkee, India
Bachelor of Technology in Computer Science and Engineering 2024 – 2028

PROFESSIONAL EXPERIENCE

Freelance Full-Stack Developer Jan 2024 – Present
Client & Independent Projects Roorkee, India

- Engineered web platforms (e.g., SevaSetu) using **React.js** and REST APIs, translating specifications into production-ready UIs.
- Architected interactive 3D WebGL websites utilizing **Three.js** and **GSAP**, achieving **95+ Lighthouse performance scores**.
- Implemented component lazy-loading, asset optimization pipelines, and responsive CSS layouts for client projects.

PROJECTS

ZeroLag Offline-First Collaborative Platform | [Live](#) | [GitHub](#)

- Architected a local-first task PWA using **React 19**, **TypeScript**, and **RxDB (IndexedDB)**, achieving **0ms UI latency** without server bottleneck.
- Built an event-sourced background sync engine with **Supabase PostgreSQL** and **Realtime WebSockets** for multi-user live presence updates.
- Integrated **Gemini Vision AI** via Vercel Serverless proxy to parse whiteboard photos into tasks, secured with **Clerk JWT + PostgreSQL RLS**.

BioSecure AI Full-Stack Biometric Platform | [GitHub](#)

- Deployed **InsightFace ArcFace 512D** face embeddings paired with **Supabase pgvector** similarity search in PostgreSQL at **90%+ accuracy**.
- Engineered a 3D Pose-Gated EWMA Embedding Drift Engine ($\alpha = 0.30$) to monitor template aging and mitigate biometric degradation.
- Built a production-ready **Flask** web app featuring a dark glassmorphic UI, automated SMTP alerts, and CSV report generation.

AuraMeet (BaatCheet) Real-Time Video Chat App | [Live](#) | [GitHub](#)

- Architected a peer-to-peer video streaming app using **Node.js**, **WebRTC**, and **Redis** delivering **sub-100ms low-latency** communication.
- Designed an ephemeral data lifecycle using **Redis pub/sub** that automatically purges chat logs and room state upon disconnect.
- Programmed a dynamic video canvas grid to optimize screen rendering performance across desktop and mobile devices.

OPEN SOURCE CONTRIBUTIONS

Core Contributor Apr 2026 – Present
Nexus Spring of Code (NSoC '26) & GSSoC Global Rank #70

- Achieved **Global Rank #70** by merging **13 Pull Requests** across 5 open-source repositories, improving security and performance.
- Migrated legacy authentication handlers to secure, rotating **HttpOnly cookie sessions** and tightened CORS security policies.
- Enhanced frontend UX responsiveness by implementing server-side pagination algorithms, debounced search inputs, and loading skeletons.